

SPM - Assignment 02

Software Project Management

Assignment 02 — Resource, Risk & Team Management

Project Title

Cloud-Based Student Assignment Submission and Management System

Group Information

Name	Roll No	Email
Uzair Ahmad	913	ozyr.uc@gmail.com
Fayaz Ahmad	924	fayazahmaddev@gmail.com

Part A — Resource Planning

1. Human Resources

Resource	Responsibilities
Project Manager	Planning, scheduling, monitoring progress
System Analyst	Requirement gathering and analysis
UI/UX Designer	Interface and user experience design
Frontend Developer	Develop client-side application
Backend Developer	Develop server-side logic and APIs
Database Administrator	Database design and management
QA/Test Engineer	Testing and bug reporting
Documentation Manager	Prepare reports and documentation

2. Technical Resources

Resource	Purpose
Laptop/Computer	Development and testing
Internet Connection	Communication and cloud access
Cloud Platform	Hosting and storage
Git & GitHub	Version control
VS Code / IntelliJ	Code development
MySQL / PostgreSQL	Database management
Browser Testing Tools	Testing compatibility
Communication Tools	Team coordination

3. Critical/Scarce Resources

Resource	Reason
Stable Internet Connection	Required for cloud access and collaboration
Cloud Hosting Credits	Limited availability for deployment
Experienced Backend Developer	Critical for system integration
Testing Devices	Limited systems for testing

Part B — RACI Matrix & Team Structure

1. RACI Matrix

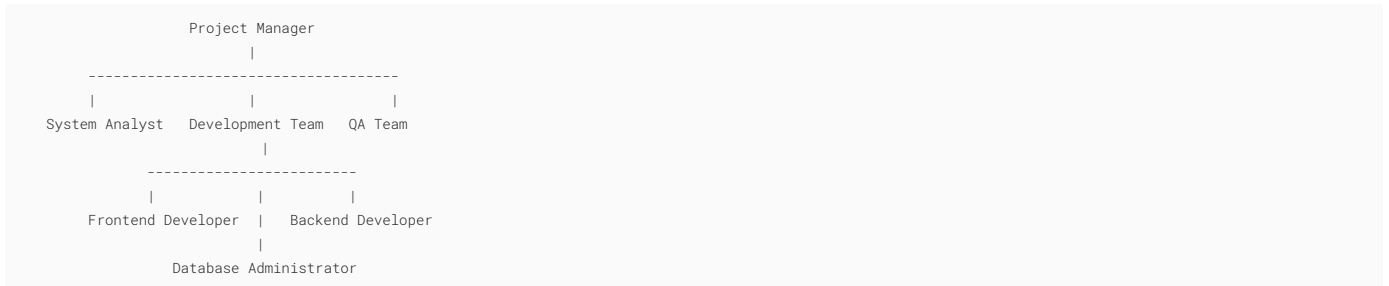
- **R** = Responsible
- **A** = Accountable
- **C** = Consulted
- **I** = Informed

Activity	Project Manager	Analyst	Developer	Tester	Documentation Manager
Requirement Gathering	A	R	C	I	I

Activity	Project Manager	Analyst	Developer	Tester	Documentation Manager
Project Planning	A	C	I	I	I
System Design	C	A	R	I	I
Development	I	C	A/R	I	I
Testing	I	I	C	A/R	I
Documentation	I	I	C	C	A/R
Deployment	A	I	R	C	I

2. Team Structure

Hierarchical Team Structure



3. Communication Flow

Communication Type	Participants	Frequency
Team Meetings	All Members	Weekly
Progress Updates	Developers → Project Manager	Daily
Requirement Discussion	Analyst ↔ Client/Instructor	As Needed
Bug Reporting	Tester ↔ Developers	Daily During Testing
Documentation Review	Documentation Manager ↔ Team	Weekly

Part C — Risk Management

1. Project Risks

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy
R1	Requirement changes during development	Medium	High	Freeze requirements after approval
R2	Internet connectivity issues	Medium	Medium	Use backup internet connection
R3	Team member absence	Medium	High	Share knowledge among team
R4	Data loss	Low	High	Regular backups and version control
R5	Delays in development	High	High	Weekly progress monitoring
R6	Cloud service downtime	Low	Medium	Choose reliable cloud provider
R7	Security vulnerabilities	Medium	High	Apply authentication and encryption
R8	Lack of communication among team	Medium	Medium	Conduct regular meetings

2. Risk Analysis Summary

- High-impact risks mainly relate to development delays and security issues.
- Medium risks involve communication and resource availability.
- Preventive planning and regular coordination can reduce most project risks.

Part D — Conflict & Coordination

1. Possible Conflict

A communication conflict may occur between frontend and backend developers due to unclear API requirements and integration delays.

2. Practical Solution

- Conduct regular coordination meetings
- Maintain proper API documentation
- Use GitHub issues/project boards for task tracking
- Clearly define responsibilities before development begins